Chapter 1

Shared Pursuit-Evasion Benchmark Formulation

1.1 Introduction

The aim of this chapter is to develop the benchmark that serves as the common experimental platform for the entire thesis. The benchmark consists of four components, each will be specified precisely before any controller is designed or any experiment is run. Section 1.2 defines the discrete-time state-space model in relative coordinates, states the modeling assumptions explicitly, and derives the benchmark dynamics matrices. Next, Section 1.3 defines the capture condition and initial condition distribution that governs the episode structure. Then, Section 1.4 states both the game-theoretic and the reinforcement learning problem formulations on this shared dynamics model, including the cost functions that define each controller's objectives. Section 1.5 defines the metrics for the experiment. Section 1.6 then defines the four experimental scenarios that generate the results of Chapter 6. In Section 1.7, the complete benchmark specification is collected into a single reference table, and concludes with Section 1.8 summarizing the chapter.

The organizing principle considered throughout this chapter is that every design choice must be made at the benchmark level, before any controller is introduced. This is intended to ensure that the comparison in Chapter 6 is an accurate comparison of design philosophies on a fixed problem; however, it is not a comparison of controllers on different problems.

1.2 Discrete-Time State-Space Model

1.2.1 Coordinate System and State Definition

The benchmark is formulated and is considered in *one spatial dimension*, with both agents moving along a shared axis. This is an intended scope restriction, as by eliminating lateral motion, the interaction geometry is reduced to its essential adversarial structure. That is, relative separation and relative closing rate, without introducing the additional state variables and modeling complexity of a 2D setting. Any language in Chapter 1 and Chapter 2 that refers to cut-in events, merges, or tailgating should be understood as being used to motivate the adversarial structure of the problem rather than implying a 2D model. The 1D benchmark is sufficient to produce a principled, interpretable comparison between the linear-quadratic (LQ) and deep reinforcement learning (DRL) design philosophies, and extension to 2D relative motion is identified as a direction for potential future research directions in Chapter 7.

The state is defined in *relative coordinates*, rather than tracking the absolute positions and velocities of two agents separately, the state is the difference between the evader's state and the pursuer's state. This choice is intended for three reasons. First, it reduces the state dimension to two, that is, relative position and relative velocity fully captures the interaction geometry of a 1D pursuit-evasion scenario. Second, it aligns the state variables directly with the quantities that govern capture. Third, it simplifies the capture condition to a constraint on the state vector itself.

Let the pursuer occupy scalar position $p_P \in \mathbb{R}$ with a velocity $v_P \in \mathbb{R}$, and the evader occupy scalar position $p_E \in \mathbb{R}$ with velocity $v_E \in \mathbb{R}$ all along a shared one-dimensional axis. The relative state vector is defined by:

$$\mathbf{x}_k = \begin{bmatrix} \Delta p_k \\ \Delta v_k \end{bmatrix} = \begin{bmatrix} p_{E,k} - p_{P,k} \\ v_{E,k} - v_{P,k} \end{bmatrix} \in \mathbb{R}^2, \tag{1.1}$$

where $k = 0, 1, \ldots, N$ denotes the discrete-time index and N is the episode horizon. The first component, Δp_k is the signed relative separation; more specifically, positive when the evader is ahead of the pursuer and negative when the pursuer has overshot. The second component, Δv_k, denotes the relative velocity that is positive when the gap is increasing and negative when the pursuer is decreasing. This sign convention is used and adopted throughout the chapter.

1.2.2 Continuous-Time Dynamics and Discretizations

Each agent is modeled as a double integrator, which means that its acceleration is the directly controlled input, and position and velocity change according to Newton's Law [1, 2]. The double integrator is a standard benchmark plant in control systems research; more specifically, it is the simplest model that captures the inertial dynamics of a physical agent whose position is governed by velocity and whose velocity is governed by applied force or acceleration. For the pursuer with control input $u_P \in \mathbb{R}$, and the evader with control input $u_E \in \mathbb{R}$, defined by:

$$\dot{p}_P = v_P, \quad \dot{v}_P = u_P, \tag{1.2}$$

$$\dot{p}_E = v_E, \quad \dot{v}_E = u_E. \tag{1.3}$$

By subtracting (1.2) from (1.3), we get the following relative dynamics:

$$\Delta\dot{p} = \Delta v, \tag{1.4}$$

$$\Delta\dot{v} = u_E - u_P. \tag{1.5}$$

In matrix form, defining $\mathbf{x} = [\Delta p,\, \Delta v]^\top$,

$$\dot{\mathbf{x}} = A_c\mathbf{x} + B_{1,c}u_P + B_{2,c}u_E, \qquad A_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B_{1,c} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \qquad B_{2,c} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{1.6}$$

The continuous-time model (1.6) is discretized using zero-order hold (ZOH) with sampling period $T_s > 0$ [1, 3]. The Zero-Order Hold (ZOH) is the standard discretization method for sampled-data control systems and is consistent with the manner in which a DRL agent its environment. More specifically, at each timestep k, the agent observes the state $\mathbf{x}_k$, selects an action, and holds that action constants for one period T_s before the next observation.

The Zero-Order Hold input matrices are obtained by integrating the matrix exponential .For the double integrator, this evaluates in closed form via the Cayley-Hamiltonian Theorem [4] (since $\mathbf{A}_c^2 = 0$, the series will truncate after two terms):

$$\mathbf{A} = e^{\mathbf{A}_c T_s} = \mathbf{I} + \mathbf{A}_c T_s = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}. \tag{1.7}$$

The Zero-Order Hold input matrices are obtained by integrating the matrix exponential against each input direction. Since $\mathbf{A}_c$ is singular, the standard formula $\mathbf{A}_c^{-1}(e^{\mathbf{A}_c T_s} - \mathbf{I})\mathbf{B}_c$

does not apply. The correct expression is given by:

$$\mathbf{B}_d = \int_0^{T_s} e^{\mathbf{A}_c \tau} \mathbf{B}_c \, d\tau \tag{1.8}$$

If we evaluate (1.8) explicitly for each input channel using

$$e^{\mathbf{A}_c \tau} = \mathbf{I} + \mathbf{A}_c \tau = \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} :$$

$$\mathbf{B}_1 = \int_0^{T_s} \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} d\tau = \int_0^{T_s} \begin{bmatrix} -\tau \\ -1 \end{bmatrix} d\tau = \begin{bmatrix} -T_s^2/2 \\ -T_s \end{bmatrix}, \tag{1.9}$$

$$\mathbf{B}_2 = \int_0^{T_s} \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau = \int_0^{T_s} \begin{bmatrix} \tau \\ 1 \end{bmatrix} d\tau = \begin{bmatrix} T_s^2/2 \\ T_s \end{bmatrix}. \tag{1.10}$$

Thus, the nominal discrete-time state equation is then:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}_1 u_{P,k} + \mathbf{B}_2 u_{E,k}, \tag{1.11}$$

where $u_{P,k} \in \mathbb{R}$ denotes the pursuer's control input at time k, and $u_{E,k} \in \mathbb{R}$ denotes the evader's control input at time k. For completeness, the dimensions of all objects in the benchmark model are collected here and are as follows: $\mathbf{x}_k \in \mathbb{R}$; $u_{P,k}, u_{E,k} \in \mathbb{R}$; $\mathbf{A} \in \mathbb{R}^{2\times 2}$, $\mathbf{B}_1, \mathbf{B}_2 \in \mathbb{R}^{2\times 1}$; $\mathbf{Q}, \mathbf{P}_N \in \mathbb{R}^{\not{2}\times\not{2}}$; and $R_1, R_2 \in \mathbb{R}_{>0}$. All the cost matrices are symmetric.

1.2.3 Modeling Assumptions

The following assumptions are made explicitly and these assumptions combine standard benchmark conventions from sampled-data control and linear-quadratic pursuit-evasion theory with deliberate scope restrictions introduced to isolate the controller-comparison question. Their purpose is not to claim a fully realistic traffic model, but rather, to define a low-dimensional, reproducible nominal benchmark on which both the model-based and learning-based controllers can be evaluated under identical conditions. These assumptions are as follows:

- **Assumption 3.1 (One-Dimensional Motion)**: Both agents move along a shared axis; more specifically, lateral dynamics, road geometry, and multi-lane interactions are not modeled. This restriction is intended as the purpose of the benchmark is to isolate the essential adversarial geometry of pursuit and escape; that is, relative separation

and relative closing rate, are without introducing additional state variables, scenario definitions, and modeling ambiguities that are associated with full 2D traffic model. In this sense, the benchmark should be read as a controlled micro-scenario that is motivated by transportation interaction, not as a complete highway simulator [5, 6, 7, 8, 9, 10, 11].

- **Assumption 3.2 (Double Integrator Plant)**: Each agent is a pure double integrator; more specifically, the acceleration is the controlled input, and there is no drag, friction, or actuator lag in the nominal model. The double integrator is a standard benchmark plant in control, since it is the simplest model that still preserves the inertial motion and produces nontrivial position, or velocity dynamics. It is therefore more appropriate in this scenario as a common plant for both the LQ game derivation and the DRL environment. It is possible for more detailed vehicle models; however, introducing them here at this stage would reduce the analytical transparency and make derivations between the two controllers harder to interpret [1, 2, 10, 9].

- **Assumption 3.3 (Perfect State Observation)**: Both controllers observe the full relative state $x_k = [\Delta p_k, \Delta v_k]^\top$ at every time step. There is no measurement noise or partial observability in the nominal setting. This assumption is used so that the benchmark isolates controller-design differences rather than estimator-design differences. If noisy or partial observations were introduced at the nominal level, then any performance gap could reflect state-estimation quality as much as control quality. Partial observability is critical in practical autonomous systems and pursuit-evasion settings; however, it is reserved here as a direction for a potential research direction rather than compressed into the baseline benchmark [5, 6, 12, 13].

- **Assumption 3.4 (Unconstrained Inputs, nominal)**: In the nominal case, both $u_{P,k}$ and $u_{E,k}$ are unconstrained real-valued scalars. This assumption is required by the nominal LQ dynamic-game formulation used in Chapter 4, where the Riccati-based saddle-point solution is derived for the unconstrained benchmark. It also creates a clean reference point for the saturation experiment of Section 3.6, in which actuator limits are introduced explicitly as a structured violation of the nominal assumptions. In other words, saturation is not ignored as it is postponed so that its effect can be measured directly rather than hidden inside the baseline design [9, 10, 14, 15, 16].

- **Assumption 3.5 (Known, time-invariant matrices)**: The matrices $\mathbf{A}, \mathbf{B}_1, \mathbf{B}_2$ are known exactly at design time and do not change during an episode. This assumption is crucial for the model-based aspect of the benchmark, since the finite-horizon LQ

controller requires explicit system matrices to calculate the backward Riccati recursion and the associated feedback gains. It also provides the nominal reference against which the model-mismatch experiment is defined. More specifically, the evaluation dynamics can later be perturbed precisely, since the benchmark begins from a fixed design-time model. The DRL pursuer is not given these matrices directly, but the benchmark itself must still specify them exactly so that both controllers operate on the same underlying plant [9, 10, 17, 18, 19].

Considering all these assumptions together, they define a benchmark that is intentionally narrow, but is methodologically useful. The benchmark is low-dimensional enough to admit an exact analytical game-theoretic solution, yet rich enough to support nontrivial comparison under model mismatch, process noise, and input saturation. This controlled structure is central to this thesis; more specifically, the objective is not model every feature of real traffic interaction at once, but to create a shared testbed on which differences between model-based and learning-based controller design can be interpreted without ambiguity [20, 21, 14, 15].

1.2.4 Nominal Parameter Values

Let $T_s = 0.1$ seconds. The sampling period is set to $T_s = 0.1s$, which is consistent with the control frequencies used in automotive and UAV control applications within the literature [20, 22] and is small enough that the Zero-Order Hold (ZOH) approximation introduces negligible error relative to continuous-time dynamics [3]. With these values, the nominal matrices evaluate to:

$$\mathbf{A} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B}_1 = \begin{bmatrix} -0.005 \\ -0.1 \end{bmatrix}, \qquad \mathbf{B}_2 = \begin{bmatrix} 0.005 \\ 0.1 \end{bmatrix} \tag{1.12}$$

The episode horizon is $N = 50$ time steps, which corresponds to 5 seconds of simulated interaction. This is long enough for the pursuer to achieve capture from a typical initial condition under the nominal LQ controller, and long enough for the DRL agent to observe a meaningful trajectory, but is short enough such that a large number of training episodes can be completed efficiently.

1.3 Capture Condition and Episode Structure

1.3.1 Capture Condition

Capture is declared when the relative separation falls below a threshold given by $\epsilon_p > 0$ and the relative velocity falls below a threshold that is given by $\epsilon_v \geq 0$ simultaneously [23, 24]. Then:

$$\mathcal{C} = \{\mathbf{x} \in \mathbb{R}^2 : |\Delta p| \leq \epsilon_p \text{ and } |\Delta v| \leq \epsilon_v\}. \tag{1.13}$$

Capture is declared at the first time step k^* such that $\mathbf{x_{k^*}}$; moreover, then the episode terminates immediately. If $\mathbf{x}_k \notin \mathcal{C}$ for all $k = 0, \ldots, N$, then the episode terminates at the horizon and capture is recorded as unsuccessful.

The two-threshold formulation (1.13) is an intended choice over a simple position-only threshold. In regards to transportation safety, the dangerous condition is not merely small separation but small separation combined with rapid closure. More specifically, a vehicle very close to another but moving away is not a safety threat, while a vehicle closing rapidly from a moderate distance is. The velocity threshold ϵ_v encodes this physical intuition directly into the capture geometry.

The nominal threshold values are $\epsilon_p = 0.5$ meters and $\epsilon_v = 1.0$ m/s. These values are held constant throughout all experimental scenarios.

1.3.2 Initial Condition Distribution

At the start of each episode, the initial state $\mathbf{x}_0$ is drawn uniformly from a rectangular region. This is given by:

$$\mathbf{x}_0 \sim \mathcal{U}\big([\Delta p_{min}, \Delta p_{max}] \times [\Delta v_{min}, \Delta v_{max}]\big), \tag{1.14}$$

with nominal values $\Delta p_{min} = 2.0$ m, $\Delta p_{max} = 10.0$ m, $\Delta v_{min} = -2.0$ m/s, $\Delta v_{max} = 2.0$ m/s. The initial relative separation is always positive, which places the evader ahead of the pursuer at episode start. The initial relative velocity spans closing and opening conditions with equal probability.

Using a distribution rather than a fixed initial conditions serves two purposes. First, it produces a more informative estimate of controller capability, since the performance measured across a range of initial geometries is more representative than performance demonstrated from a single starting state. Second, it ensures that the DRL pursuer is trained and evaluated over a variety of encounter conditions, reducing the risk that the learned policy is overfitted to one particular initial geometry.

The same initial condition distribution is used for LQ evaluation, DRL training, and DRL

evaluation. This is a strict requirement of the benchmark design; that is, any difference in performance between the two controllers must be attributable to the controllers, not to differences in the initial condition regime.

1.4 Problem Formulation

The model-based controller is derived from the following two-player, zero-sum, finite-horizon LQ dynamic game. Let $u_{P,k}$ be the pursuer's control input (or Player 1), and $u_{E,k}$ be the evader's control input (or Player 2) at time step k. Now, let us define the finite-horizon quadratic cost functional:

$$J = \mathbf{x}_N^\top \mathbf{P}_N \mathbf{x}_N + \sum_{k=0}^{N-1} \left[\mathbf{x}_k^\top \mathbf{Q} \mathbf{x}_k + u_{P,k}^\top R_1 u_{P,k} - u_{E,k}^\top R_2 u_{E,k} \right], \tag{1.15}$$

where $\mathbf{Q} \in \mathbb{R}^{2\times 2}$ denotes the state-cost matrix, $R_1 > 0$ denotes the pursuer's control-cost weight, $R_2 > 0$ is the evader's control-cost weight, and $P_N \in \mathbb{R}^{2\times 2}$ is the terminal state-cost matrix. The sign convention in (1.15) reflects the zero-sum structure. More specifically, the pursuer minimizes J, while the evader maximizes J. The pursuer aims to drive the state toward zero or *capture*, while the evader aims to maintain the state away from zero or *capture*. Both players are penalized for large control effort through the quadratic terms $R_1 u_{P,k}^2$ and $R_2 u_{E,k}^2$, which prevents degenerate bang-bang solutions and produces smooth, interpretable trajectories.

The two-plater zero-sum game (1.15) is subject to (1.11) is solved for a saddle-point equilibrium [9]; that is, a pair of policies (π_P^*, π_E^*) such that:

$$J\left(\pi_P^*, \pi_E\right) \leq J\left(\pi_P^*, \pi_E^*\right) \leq J\left(\pi_P, \pi_E^*\right) \tag{1.16}$$

Since the game is zero-sum, the saddle-point equilibrium coincides with the Nash equilibrium [9]. The term **saddle-point equilibrium** is used throughout this thesis since it is the standard terminology for zero-sum games and captures the minimax structure more precisely than the general Nash concept.

The cost-matrix values that are used in this thesis are given as follows:

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 0.1 \end{bmatrix}, \qquad R_1 = 0.01, \qquad R_2 = 0.02 \qquad \mathbf{P}_N = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix} \tag{1.17}$$

The relative weighting in $\mathbf{Q}$ is used to penalize relative position errors more heavily than

relative velocity errors, which reflects the physical priority of closing the gap. The slight asymmetry between the control costs is intentional rather than cosmetic. The benchmark keeps the pursuer effort weight at $R_1 = 0.01$, and increases only the evader effort weight to $R_2 = 0.02$; which is the smallest correction needed to preserve a well-posed finite-horizon saddle-point benchmark while leaving the rest of the problem unchanged, as will be shown in Chapter 4. The resulting weights still allow for aggressive nominal inputs, which keep the saturation experiment meaningful.

1.4.1 Reinforcement Learning Problem Statement

The learning-based pursuer is trained within a Markov Decision Process (MDP) that is defined on the same state and action spaces as the LQ game [25]. The MDP is specified by the tuple (S, A, T, r, γ):

- **State Space** ($S = \mathbb{R}^2$): The relative state $\mathbf{x}_k = \left[\Delta p_k, \Delta v_k\right]^\top$ is the full observation provided to the pursuer at each time step and no additional information is given.

- **Action Space** A: The pursuer selects a scalar acceleration $u_{P,k}$ at each step, and in the nominal benchmark, no hard physical actuator limit is imposed. In implementation, the DRL policy-network output is numerically bounded through a scaled tanh squashing function with a large symmetric interval $[-u_{clip}, u_{clip}]$ is chosen so that the bound remains inactive under nominal benchmark trajectories; moreover, the effective action space is therefore unconstrained for practical purposes during nominal training. In the saturation scenario (Section 3.6), a hard post-network clip is applied to the pursuer only, enforcing $u_{P,k} \in [-u_{clip}, u_{clip}]$ for both the LQ pursuer and the frozen DRL pursuer at evaluation time. The evader's action $u_{E,k}$ remains the fixed saddle-point equilibrium strategy from the LQ game and is not saturated.

- **Transition Function** T: The state transitions according to (1.11), with additive noise $\mathbf{w}_k$ present in the noise experiment and absent otherwise.

- **Reward Function** r: Will be defined in Section 3.4.3 below.

- **Discount Factor** $\gamma \in (0, 1]$: Set $\gamma = 1.0$ for the nominal and perturbation experiments, making the DRL objective equal to the undiscounted finite-horizon cost, consistent with the LQ game formulation.

The DRL pursuer's objective is to find a policy $\pi_P : S \to A$ that maximizes expected cumulative reward is defined by:

$$\pi_P^* = \arg\max_{\pi_P} \mathbb{E}\left[\sum_{k=0}^{N-1} \gamma^k r(\mathbf{x}_k, u_{P,k})\right] \tag{1.18}$$

It is important to note what the DRL pursuer does *not* receive; that is, the system matrices $\mathbf{A}, \mathbf{B}_1$, and $\mathbf{B}_2$, the cost matrices $\mathbf{Q}, \mathbf{R}_1, \mathbf{R}_2, \mathbf{P}_N$, or the evader's policy. The pursuer learns entirely from the scalar reward signal and the observed state transitions.

1.4.2 Reward Function Design

The reward function is designed to align the DRL pursuer's learning objective as closely as possible with the LQ game's cost structure, so that any performance gap between the two methods that can be interpreted as a consequence of design philosophy rather than as an artifact of incompatible objectives [20, 26]. The benchmark therefore preserves the two engineering priorities that most directly below to the pursuer; that is reducing the relative state and avoiding wasteful control effort. The reward alignment between the LQ cost (1.15) and the DRL objective (1.18) is achieved by the following reward signal given by:

$$r(\mathbf{x}_k, u_{P,k}) = \begin{cases} r_{\text{capture}}, & \text{if } \mathbf{x}_k \in \mathcal{C}, \\ -\left(\mathbf{x}_k^\top Q \mathbf{x}_k + R_1 u_{P,k}^2\right), & \text{otherwise}, \end{cases} \tag{1.19}$$

where $r_{capture} > 0$ is a positive bonus awarded upon capture, and the step-level penalty reproduces the pursuer-side running cost from the LQ benchmark. The capture bonus is set to $r_{capture} = 100$, which is large enough relative to the per-step penalties to ensure that capture is strongly preferred over the accumulation of small intermediate rewards and since the episode capture, the bonus is received at most once per episode.

Let us now justify our design decisions in (1.19). Using the same $\mathbf{Q}$ and $\mathbf{R_1}$ matrices in the reward as in the LQ cost (1.15), and preserves the same state-and-effort tradeoff that governs the model-based pursuer. This is important, since if the DRL pursuer were trained under materially different weights, then any later difference in performance could be attributed either to the controller architecture or to a different underlying objective. By keeping the state penalty and pursuer-effort penalty identical, and the benchmark makes the comparison interpretable.

The LQ cost in (1.15) also includes the evader-effort term given by $-R_2 u_{E,k}^2$, which represents the evader's contribution to the zero-sum game value. This term is excluded from the DRL reward because the pursuer is not given the evader's action as part of its observation and therefore cannot evaluate that quantity directly during learning. More importantly, it

includes an inferred or estimated evader-effort term would move the benchmark away from the intended comparison and toward a different problem involving opponent modeling inside the reward itself. The benchmark instead keeps the DRL signal tied to quantities the pursuer actually experiences. More specifically, the observed relative state, its own control effort, and whether capture has occurred.

The terminal cost $\mathbf{x}_N^\top \mathbf{P}_N \mathbf{x}_N$ is also excluded from the stepwise reward and this is a deliberate simplification rather than an oversight. In the finite-horizon LQ game, the terminal term emphasizes final-state proximity. More specifically, in the DRL environment, that role is served in practice by the accumulated state penalty together with the one-time capture bonus. If we add a separate terminal shaping term, it would complicate the software path without materially changing the basic incentives of the benchmark.

The consequence is that the DRL reward is aligned with, however not mathematically identical to, the full two-player game value in (1.15). As a result, that limitation should be read carefully. It does not invalidate the comparison, since the main benchmark ingredients remain shared. More specifically, identical dynamics, identical state variables, identical capture logic, identical pursuer running-cost weights, and a fixed Nash evader. Later in Chapter 6, it will interpret any residual gap between the two controllers as a combination of finite-sample learning effects, function-approximation error, and the controlled objective asymmetry introduced here, rather than as pure controller quality in the abstract.

1.5 Evaluation Metrics

We define five metrics to be used to evaluate and compare the two controllers. All metrics are computed over $M = 500$ evaluation episodes per experimental condition, with initial conditions drawn independently from the distribution (1.14). This episode count provides a stable empirical estimate of capture rate and other per-episode statistics while remaining computationally tractable; that is confidence intervals are reported alongside all point estimates to characterize estimation uncertainty, these are as follows:

- **Metric 1** (Capture Rate ($\hat{p}$)): The fraction of evaluation episodes in which capture is achieved before the horizon:

$$\hat{p} = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1} \left[\exists\, k \leq N : \mathbf{x}_k^{(m)} \in \mathcal{C} \right]. \tag{1.20}$$

  This is the primary metric. A 95% Wilson confidence interval [27] is reported alongside the point estimate.

- **Metric 2** (Mean Capture Time ($\bar{k}^*$)): The mean time step at which capture is achieved, computed over episodes where capture over all episodes:

$$\bar{k}^* = \frac{1}{|\mathcal{M}_c|} \sum_{m \in \mathcal{M}_c} k_m^*, \tag{1.21}$$

  where $\mathcal{M}_c$ is the set of episodes in which capture occurred and k_m^* is the capture time step in episode m. It is important to note that lower is better from the pursuer's perspective.

- **Metric 3** (Cumulative State-and-Control Cost) $\bar{J}$: The mean value of the LQ cost functional (1.15) evaluated over all episodes:

$$\bar{J} = \frac{1}{M} \sum_{m=1}^{M} \left[\mathbf{x}_N^{(m)\top} P_N \mathbf{x}_N^{(m)} + \sum_{k=0}^{N-1} \left(\mathbf{x}_k^{(m)\top} Q \mathbf{x}_k^{(m)} + R_1 \big(u_{P,k}^{(m)}\big)^2 \right) \right]. \tag{1.22}$$

  This metric is evaluated by using the LQ cost matrices for both controllers, thus it measures both controllers against the same objective function and lower is better.

- **Metric 4** (Total Control Effort) $\bar{E}$: The mean sum of squared pursuer control inputs across an episode:

$$\bar{E} = \frac{1}{M} \sum_{m=1}^{M} \sum_{k=0}^{N-1} \big(u_{P,k}^{(m)}\big)^2. \tag{1.23}$$

  This metric is reported separately from Metric 3 to reveal whether performance differences are associated with differences in aggressiveness of control.

- **Metric 5** (Sample Efficiency – DRL only): The learning curve of the DRL pursuer, which is expressed as mean episodic reward versus training episode count, is averaged across five independent random seeds. Then, sample efficiency is measured by the episode count $\hat{n}_{0.9}$ at which the rolling mean episodic reward (where the window is 100 episodes) first reaches 90% of its final converged value and remains above that threshold for at least 50 consecutive episodes. This metric has no LQ analog, as the LQ controller requires zero environment interactions to produce its policy. It is therefore reported descriptively rather than comparatively; more specifically, it characterizes the training burden of the DRL design philosophy and the practical cost of achieving a given level of pursuit performance through learning alone. Later in Chapter 6, this metric will be used separately from the head-to-head controller comparisons (Metrics 1-4) and is not used to rank the two controllers against each other.

1.6 Experimental Scenarios

Four specific experimental scenarios are designed to answer the four research questions established in Chapter 1. In all scenarios, the evader plays the LQ saddle-point equilibrium strategy will be derived in Chapter 4. All other scenario parameters are as specified in the preceding sections unless explicitly stated otherwise.

1.6.1 Scenario 1: Nominal Performance (RQ1 and RQ2)

In Scenario 1, both controllers operate on the nominal dynamics (1.11) with the nominal matrices (1.12), no additive noise, and no input constraints. This is the cleanest possible comparison, as the LQ controller is evaluated on exactly the dynamics on which it was trained.

Scenario 1 generates the primary performance comparison (given by Metrics 1-4) and the DRL learning curve (Metric 5). Hence, the results from this experimental scenario answers Research Question 1 (i.e., how closely can DRL match the LQ equilibrium, and at what sample cost?) and Research Question 2 (i.e., How do the two methods compare across each performance dimension individually?).

1.6.2 Scenario 2: Model Mismatch (RQ3)

Scenario 2 isolates sensitivity to design-time model error. Both controllers are designed once under the nominal benchmark, and both are then evaluated on the same perturbed simulation dynamics. The point of the scenario is therefore not to ask how well either method could perform after being retuned for the true plant, but how robust each nominally designed controller remains when the plant departs from the assumptions used to produce it.

Mismatch is introduced by perturbing the sampling period T_s is used to generate the simulation matrices while holding the design-time matrices fixed at their nominal values. More specifically, the simulation uses $\tilde{T}_s = T_s(1+\delta)$ for mismatch levels $\delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ which corresponds to 10% through 50% deviations in the effective sampling period. This parameterization is attractive for the present benchmark since it perturbs both $\tilde{A}$ an $\tilde{B}_{1,2}$ in a smooth and interpretable way while preserving the same relative-coordinate structure. In other words, the mismatch does not change the type of system that is being studied, it changes the accuracy with which the nominal design model represents that system.

The lower end of the range represents moderate design error, while the upper-end is intentionally stressful. A 10% deviation is consistent with the kind of discrepancy that

can arise from approximate system identification, imperfect calibration, or a nominal model carried across similar platforms. A 50% deviation is not meant to be routine as it is a stress test designed to make degradation visible if the controller is tightly tied to the nominal model. The intermediate values provide enough resolution to distinguish gradual degradation from abrupt failure.

For each mismatch level δ, Metrics 1-4 are computed for both controllers. The resulting performance-versus-mismatch curves answer RQ3 directly by showing not only whether performance degrades; however, how it degrades and whether the two controllers degrade in the same way.

1.6.3 Scenario 3: Additive Process Noise (RQ4a)

Scenario 3 tests robustness to stochastic perturbation of the state transition while leaving the nominal controller designs unchanged, and the benchmark dynamics become:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}_1 u_{P,k} + \mathbf{B}_2 u_{E,k} + \mathbf{w}_k, \qquad \mathbf{w}_k \sim \mathcal{N}(0, \sigma^2 I_2), \tag{1.24}$$

where σ^2 denotes the noise variance. Both controllers observe the noisy state and use it directly in the next control computation; i.e, neither is paired with a separate state estimator. This keeps the scenario focused on control robustness rather than estimator design.

Noise levels $\sigma \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$ are evaluated and these values should be interpreted relative to the benchmark state variables and capture thresholds rather than as claims about one particular sensing stack. At the lower end, $\sigma = 0.05$ produces mild perturbations compared with the position and velocity scales of the benchmark. At the higher end, $\sigma = 0.50$ is deliberately large relative to both the position threshold $\epsilon_p = 0.5$ m and the velocity threshold $\epsilon_v = 1.0$ m/s, so that the noise directly competes with the capture condition itself. The sweep therefore spans a meaningful range from modest stochastic disturbance to a regime in which state perturbation materially changes the outcome of an episode.

For each noise level, Metrics 1-4 are computed for both controllers. The resulting curves reveal whether either design philosophy is more resilient to stochastic state perturbation and whether nominal analytical optimality remains informative once the deterministic benchmark assumptions are weakened.

1.6.4 Scenario 4: Input Saturation (RQ4b)

Scenario 4 probes the benchmark's most explicit structural asymmetry; more specifically, the LQ controller is derived under an unconstrained-input assumption, whereas the evaluation environment can impose a hard magnitude limit on the pursuer's action. The pursuer input is therefore clipped according to:

$$u_{P,k}^{\text{sat}} \leftarrow \text{clip}(u_{P,k}, -u_{\max}, u_{\max}), \tag{1.25}$$

where $u_{\max} > 0$ is the saturation limit and $\text{clip}(a, b, c) = \max(b, \min(a, c))$. The evader's control is not saturated in any scenario and this is a deliberate choice that places the adversarial pressure on the pursuer.

The LQ controller computes its unconstrained saddle-point equilibrium policy and then applies clipping (1.25) post-hoc, with no modification to the Riccati recursion. This is the natural failure mode of a model-based controller derived under Assumption 3.4. More specifically, when the unconstrained solution exceeds the physical limit, it is truncated without any compensating adjustment. The DRL pursuer, on the other hand, was trained with its actions that are passed through the same clip operation (1.25), so it is has had the opportunity to learn a policy that accounts for the constraint.

Saturation levels $u_{\max} \in \{0.5, 1.0, 2.0, 5.0, 10.0\}$ m/s^2 are evaluated and at $u_{\max} = 10.0$ m/s^2, the constraint is rarely active under nominal conditions and the results should approach the nominal-case comparison from Scenario 1. At $u_{\max} = 0.5$ m/s^2, clipping is frequent and the effect of actuator limitation becomes impossible to ignore. The intermediate values bridge these two regimes and make it possible to observe whether performance degrades smoothly or undergoes a sharper change once the unconstrained policy begins to collide with the actuator boundary.

The range is grounded in representative longitudinal actuation scales from automated-vehicle and related control literature; however its role in this thesis is primarily comparative rather than descriptive [20, 22, 28]. The key point is that the sweep moves from near-nominal to heavily constrained operation while keeping ever other benchmark ingredient fixed. As a result, this makes scenario 4 the clearest direct test of whether the nominal LQ design and the nominal DRL design fail in the same way once actuator limits are imposed.

1.7 Benchmark Summary and Table

Table 1.1 collects the complete benchmark specification and summarizes it into one table. Every quantity that is listed here is fixed before any controller is introduced and remains

Table 1.1: Complete benchmark specification. Parameters marked with † vary across experimental conditions as specified in Section 3.6.

Parameter	Symbol	Value
Sampling period	T_s	0.1 s
State matrix	A	$\begin{bmatrix} 1 & 0.1 \\ 0 & 1 \end{bmatrix}$
Pursuer input matrix	B_1	$\begin{bmatrix} -0.005 \\ -0.1 \end{bmatrix}$
Evader input matrix	B_2	$\begin{bmatrix} 0.005 \\ 0.1 \end{bmatrix}$
Horizon	N	50 steps (5 s)
Position capture threshold	ϵ_p	0.5 m
Velocity capture threshold	ϵ_v	1.0 m/s
Initial position range	$[\Delta p_{\min}, \Delta p_{\max}]$	$[2.0, 10.0]$ m
Initial velocity range	$[\Delta v_{\min}, \Delta v_{\max}]$	$[-2.0, 2.0]$ m/s
State cost matrix	Q	$\begin{bmatrix} 1 & 0 \\ 0 & 0.1 \end{bmatrix}$
Pursuer control cost	R_1	0.01
Evader control cost	R_2	0.02
Terminal cost matrix	P_N	$\begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}$
Capture bonus (DRL)	r_{capture}	100
Discount factor (DRL)	γ	1.0
Episodes per condition	M	500
DRL training seeds	—	5
Mismatch level†	δ	$\{0.1, 0.2, 0.3, 0.4, 0.5\}$
Noise std. dev.†	σ	$\{0.05, 0.10, 0.20, 0.30, 0.50\}$
Saturation limit†	$u_{\max}$	$\{0.5, 1.0, 2.0, 5.0, 10.0\}$ m/s^2

constant across both controllers and all experimental scenarios unless explicitly identified as a varying parameter in that scenario's definition. Table 1.1 is not intended as an appendix-style convenience; moreover, it is a part of the benchmark's methodological role. By gathering the full specification in one place, it makes clear that the comparison in later chapters is not between differently tuned problems, but it is between two controller-design philosophies applied to one common problem statement.

1.8 Chapter Summary

The aim of this chapter was to establish the benchmark that supports the remainder of the thesis. The relative-coordinate double-integrator model supplies a low-dimensional but

nontrivial pursuit-evasion testbed; moreover, the capture condition, initial-condition distribution, and cost structure translates that model into a finite-horizon benchmark problem. In addition, the evaluation metrics and scenario definitions that specify exactly how the two controller designs will be compared. In summary, these elements define the common experimental language of the thesis.

More importantly, Chapter 1 allows for the interpretation of the later results. The LQ formulation and the DRL formulation do not differ, since they see different state variables, different episode structure, or different evaluation metrics. They differ because one is derived from explicit model knowledge and the other is learned from sampled interaction under an aligned pursuer-side objective. Then, the perturbation scenarios then weaken particular nominal assumptions in a controlled way; more specifically, Scenario 2 weakens the known-model assumption, Scenario 3 weakens the deterministic-transition assumption, and Scenario 4 weakens the unconstrained-input assumption. As a result, that structure is what makes the comparisons that will be in Chapter 6 will be more interpretable rather than anecdotal.

Finally, with the benchmark now fixed, Chapter 4 can now derive the exact model-based saddle-point controller and Chapter 5 will implement the learning-based pursuer on the same state, action, and episode structure. The rest of this thesis therefore will build on this single benchmark specification rather than on moving definitions.

Appendix: Chapter-Specific Sources

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